

The empirical recurrence for OEIS A297342: recovering a conjecture that is not written down

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Abstract

OEIS A297342 records “Empirical recurrence of order 82”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 113$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 82, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 82 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A297342 is “Number of $5 \times n$ 0..1 arrays with every 1 horizontally or antidiagonally adjacent to 0 or 2 neighboring 1s.”. It has offset 1 and begins

32, 144, 1847, 13428, 130297, 1091456, 9831967, 85542000, ...

The entry states:

Empirical recurrence of order 82 (see link above)

As of the “Last modified” line on the live entry (Dec 28 2017 11:35:03, revision 4) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 113$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 113$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 226$ exact terms of the model returns order 82 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{82} c_i a(n-i),$$

c_1	7	c_{22}	−47576145	c_{43}	−416530094	c_{64}	−3665655
c_2	31	c_{23}	−60012155	c_{44}	−216205430	c_{65}	2713417
c_3	−92	c_{24}	158499068	c_{45}	38818558	c_{66}	2478181
c_4	−407	c_{25}	−190115807	c_{46}	611562278	c_{67}	882982
c_5	−31	c_{26}	180137692	c_{47}	438752131	c_{68}	−10201
c_6	3556	c_{27}	−140928839	c_{48}	51169326	c_{69}	−136274
c_7	976	c_{28}	−1266841	c_{49}	−447581681	c_{70}	−75307
c_8	−15274	c_{29}	480724637	c_{50}	−584561350	c_{71}	−31470
c_9	−9468	c_{30}	−738185529	c_{51}	−218797794	c_{72}	−7350
c_{10}	72193	c_{31}	390683424	c_{52}	245419595	c_{73}	−269
c_{11}	42524	c_{32}	−183059158	c_{53}	461958181	c_{74}	2746
c_{12}	−301201	c_{33}	−494496303	c_{54}	240857245	c_{75}	1125
c_{13}	−317774	c_{34}	776031093	c_{55}	−92201912	c_{76}	266
c_{14}	1549330	c_{35}	−346215235	c_{56}	−245939715	c_{77}	12
c_{15}	−1000081	c_{36}	516672794	c_{57}	−154043837	c_{78}	−62
c_{16}	6564	c_{37}	−55791901	c_{58}	15622351	c_{79}	−1
c_{17}	−1767492	c_{38}	−350745406	c_{59}	91026599	c_{80}	−7
c_{18}	9537307	c_{39}	244784018	c_{60}	63004015	c_{81}	−1
c_{19}	−5598376	c_{40}	−327265398	c_{61}	4395658	c_{82}	1
c_{20}	−47607526	c_{41}	367618890	c_{62}	−21850536		
c_{21}	93238702	c_{42}	−128036797	c_{63}	−16654234		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 82, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $82 + 4$ terms removed returns order 82 as well, so $\deg q_{\min} = 82 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 18 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $82 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 82 that this sequence satisfies — and that family turns out to have exactly one member.

References

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